

```

        bot.sendMessage(chat_id, 'Security camera is activated.')#Put your welcome note here while true:
    main()
    bot = telepot.Bot('*****:*****_*****')# API key bot.message_loop(handle)
def main():
    global chat_id
    global motion
    global motionNew
    if GPIO.input(PIR) == 1:
        print("Motion detected")
        motion = 1
    if motionNew != motion:
        motionNew = motion
        sendNotification(motion)
    elif GPIO.input(PIR) == 0:
        print("No motion detected")
        motion = 0
    if motionNew != motion:
        motionNew = motion
    def sendNotification(motion):
        global chat_id
        if motion == 1:
            filename = "/home/pi/Desktop/" + (time.strftime("%b%d_%H%M%S")) # default folder camera.start_recording(filename + ".h264")
            sleep(5)
            camera.stop_recording()
            command = "MP4Box -add " + filename + '.h264' + " " + filename + '.mp4' # Convert .H264 to MP4 print(command)
            call([command], shell=True)
            bot.sendVideo(chat_id, video = open(filename + '.mp4', 'rb'))
            bot.sendMessage(chat_id, 'The motion sensor is triggered!') #Message displayed on Telegram screen while 1:
            time.sleep(5)

```

Fig. 9: Example code snippet for API key



Fig. 10: Author's Telegram bot Watchbot's screenshot with glimpses of video storage code

that interacts with the bot and receives commands. Open an editor on the Raspberry Pi to develop the code for connecting the Raspberry Pi to the Telegram bot. We used Python3 IDE, and testing is shared in the final part. Create a new file and name it `telegram_video.py`:

In the code, first import the required libraries, initialise the message loop, import calls, and set up the camera characteristics and GPIO pins for the sensor.

Initialise the chat ID, message text for display, and the correct key and initialisation text. You can place the welcome message and the loop for checking motion, capturing video frames, and converting them into .mp4. Fig. 9 shows an example code snippet for the API key.

EFY note. 1. Ensure correct video conversion using MP4Box.

2. Insert the `bot.send` commands to

share the video file and text message.

```

command="MP4Box -add "+filename+ '.h264'
+" "+filename+ '.mp4' # Convert .h264 to MP4
bot.sendVideo(chat_id, video =
open(filename + '.mp4', 'rb'))
bot.sendMessage(chat_id, 'The motion
sensor is triggered!')

```

Construction and testing

Connect the camera to the Raspberry Pi camera port and then connect the PIR motion sensor according to the circuit diagram given in Fig. 2.

Power the Raspberry Pi, connect it to a Wi-Fi network and the internet, and run the code. Create motion in front of the motion sensor. It will capture the video frames, convert them into .mp4, and send them to the connected Telegram bot on your phone or device. Fig. 10 shows the author's Telegram bot - Watchbot screenshot with glimpses of the video storage code. **EFY**

EFY Note

The source code of this project is available

for free download at www.electronicsforu.com

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